

# EXECUTIVE PROPOSAL: URBAN HEAT MITIGATION VIA AI/ML

*High-Novelty Model Architecture & Selection Enhancement Guide*

## Project Selection Pitch: Why This Framework Wins

Hackathon judges evaluating selection rounds look for three things: academic novelty, real-world feasibility, and rigorous mathematical architecture. This document outlines the cutting-edge architectural advancements being integrated into our Physics-Informed Neural Network (PINN) and Evolutionary Optimization pipeline to make it an elite, research-grade framework for smart-city governance.

### 1. Cutting-Edge Model Enhancements (Architectural Novelty)

#### Enhancement 1: Multi-Temporal PINN (Diurnal & Nocturnal Physics)

- Current Implementation: The V4 PINN trains on a single daytime thermal snapshot (Landsat 8 LST).
- The Novel Upgrade: We are incorporating time-dependent thermal inertia into the Surface Energy Balance (SEB) loss function ( $G = C_v * dT/dt$ ). By pairing day-night thermal rasters (MODIS/ECOSTRESS), the PINN learns how dense concrete traps heat during the day and releases it at night.
- Selection Appeal: Capturing the Nocturnal Urban Heat Island effect is the holy grail of urban climatology. Judges will instantly recognize this as a breakthrough over standard static models.

#### Enhancement 2: Empirical Blue Space Extraction (Sentinel-2 B11)

- Current Implementation: The water index (NDWI) is simulated via random distributions as a placeholder.
- The Novel Upgrade: Updating `gee_extraction.py` to extract Sentinel-2 Band 11 (Shortwave Infrared) and Band 3 (Green) to compute empirical, verified NDWI. This directly feeds into the Latent Heat Flux (LE) physics equation.
- Selection Appeal: Replaces mock proxies with satellite-verified proof of how wetlands act as thermal sinks, boosting the model R-squared accuracy score above 0.45.

#### Enhancement 3: Unsupervised Local Climate Zone (LCZ) Clustering

- Current Implementation: Spatial zoning is done via a simple median split on building heat (BAH > median).
- The Novel Upgrade: Replacing the median split with an unsupervised K-Means Machine Learning clustering engine operating on [NDVI, Albedo, BAH, LST] to automatically segment the city into official WUDAPT Local Climate Zones.
- Selection Appeal: Aligns the framework with international climatology standards (WUDAPT), demonstrating rigorous data science workflows to technical judges.

#### Enhancement 4: NSGA-III Multi-Objective Ward-Level Budgeting

- Current Implementation: NSGA-II evaluates the top 100 hotspots collectively with 2 objectives.
- The Novel Upgrade: Moving to pymoo NSGA-III to optimize 3+ competing municipal objectives per ward: (1) Minimize Surface Temp, (2) Minimize City Implementation Cost, and (3) Maximize Socio-Economic Benefit.

# EXECUTIVE PROPOSAL: URBAN HEAT MITIGATION VIA AI/ML

*High-Novelty Model Architecture & Selection Enhancement Guide*

- Selection Appeal: Proves the AI is ready for real-world smart-city governance and municipal budget deployment.

## 2. Quick Comparison: Standard Submissions vs. Our Novel Framework

| Feature Area      | Standard Hackathon Models                | Our Novel Framework (V4+)                    |
|-------------------|------------------------------------------|----------------------------------------------|
| ML Methodology    | Standard Deep Learning / Random Forests  | Physics-Informed Neural Network (PINN)       |
| Thermodynamics    | Unbounded (AI can hallucinate -50C)      | Strict Surface Energy Balance (SEB) loss     |
| Mitigation Engine | Trial & error simulation guessing        | NSGA-II/III Genetic Algorithm Pareto front   |
| Spatial Zoning    | Homogeneous flat city rules (0.14C drop) | Kundu 2026 Heterogeneous zoning (3.35C drop) |
| Municipal Reality | Ignores city budget and density limits   | Enforces ward-level 20% spending constraints |

*Document prepared for Hackathon Selection Committee · Delhi-NCR Thermal Mitigation Architecture*